

ia-26/HC.py

```
1 import numpy as np
2
3
4 # =====
5 # Hill Climbing
6 # =====
7
8 def hill_climbing(
9     objective_function,
10     initial_solution,
11     bounds,
12     max_iter=5000,
13     step_size=0.001
14 ):
15
16     # Solução inicial
17     current_solution = initial_solution.copy()
18
19     current_value = objective_function(
20         current_solution
21     )
22
23     best_solution = current_solution.copy()
24     best_value = current_value
25     history = [best_value]
26
27     # Loop principal
28     for iteration in range(max_iter):
29         # Gera vizinho
30         neighbor = (
31             current_solution
32             + np.random.normal(
33                 0,
34                 step_size,
35                 len(current_solution)
36             )
37         )
38
39         neighbor = np.clip(
40             neighbor,
41             bounds[0],
42             bounds[1]
43         )
44
45         neighbor_value = objective_function(
46             neighbor
47         )
48
49         # Aceita apenas melhoria
50         if neighbor_value < current_value:
```

```
51
52         current_solution = neighbor
53         current_value = neighbor_value
54
55         if current_value < best_value:
56
57             best_solution = (
58                 current_solution.copy()
59             )
60
61             best_value = current_value
62
63         history.append(best_value)
64
65     return (
66         best_solution,
67         best_value,
68         history
69     )
```